JESSICA DICHAKDJIAN

Robotics Researcher | Robot Learning, Manipulation & Teleoperation

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LinkedIn • GitHub • Portfolio • **Ego2Robot**

RESEARCH PROFILE

Robotics & Mechatronics MSc candidate at Politecnico di Milano, conducting thesis research at University College London (UCL) in **robot learning, manipulation, and shared autonomy**. Experience with Python/PyTorch experimentation, MuJoCo and LIBERO simulation, human demonstration retargeting, model evaluation, and physical robot teleoperation. Interested in VLA policies, imitation learning, uncertainty-aware autonomy, and translating human demonstrations into robotic execution.

TECHNICAL SKILLS

Machine Learning & Research: Python, PyTorch, NumPy, supervised learning, imitation-learning concepts, Diffusion Policy experimentation, evaluation pipelines, data analysis, Git.

Robotics & Simulation: MuJoCo, LIBERO, Franka Panda simulation, ROS Noetic, kinematics, Jacobians, trajectory planning, robot control, physics-based simulation.

Systems & Development: Linux, C/C++, MATLAB, Unity/C#, Meta Quest/XR, Arduino, Raspberry Pi, robot hardware integration, testing and debugging.

Current Research Focus: Vision-Language-Action model integration and evaluation; Gaussian Process uncertainty estimation in thesis design; human-in-the-loop learning and corrective demonstrations.

RESEARCH & SELECTED PROJECTS

Ego2Robot — Human Demonstration Retargeting for Robotic Manipulation Oct 2026

Independent Robot Learning Challenge Project | Humanoid Internship Submission *GitHub*

- Developed an end-to-end pipeline transforming **personally recorded phone manipulation videos** into sparse 2D annotations and motion priors for a simulated Franka Panda in LIBERO.
- Implemented geometric, object-relative retargeting and a phase-aware, closed-loop pick-and-place controller using human motion guidance and simulator-state metric anchoring.
- Achieved successful placements in **6/6 fixed-state demonstration runs**; evaluated human, straight, and shuffled route priors in matched simulator conditions.
- Built Python evaluation tools, annotation validation, automated tests, trajectory visualisations, and reproducible experiments; documented negative results and limitations.

Uncertainty-Aware Shared Autonomy for Robotic Manipulation Aug 2026 – Present

Visiting MSc Thesis Researcher | University College London (UCL) *Repository*

- Investigating learning-based manipulation with a **Franka Panda in MuJoCo**, building on an existing UCL robotics benchmark.
- Configured and performed preliminary **Diffusion Policy** experiments; currently working on integration and evaluation of Vision-Language-Action models.
- Designing a shared-autonomy framework combining Gaussian Process uncertainty estimation with selective human corrective demonstrations. Planned XR integration builds on previous Unity/Meta Quest teleoperation work.

XR Teleoperation of a Physical Braccio Robot MSc Project

GitHub

- Integrated Meta Quest tracking, Unity, inverse kinematics, ROS Noetic, Arduino, and a **physical robotic manipulator** into a human-controlled teleoperation system.
- Compared two control architectures; moved inverse kinematics to Unity/BioIK to shorten the communication pathway and improve responsiveness.

Industrial Robotics — Manipulation & Validation MSc Project

GitHub

- Modelled a custom 4-DOF manipulator using forward/inverse kinematics, Jacobians, workspace analysis and Simscape Multibody; programmed physical collaborative-robot manipulation and inspection workflows.

RESEARCH & ENGINEERING EXPERIENCE

CERN Summer Student — ATLAS Detector R&D Jun – Aug 2023

CERN, Geneva, Switzerland

- Performed experimental characterisation of an RPC-BIL detector prototype for the ATLAS Muon Spectrometer upgrade.
- Analysed measurements, compared experimental and simulated behaviour, and characterised signal propagation

(**20.7 cm/ns**).

EDUCATION

MSc Robotics & Mechatronics Engineering Sep 2024 – Present
Politecnico di Milano, Italy
Thesis research at University College London (UCL). Relevant study: Advanced Robotics, Multivariable Control, Real-Time Systems, Collaborative Robotics.

BSc Robotics & Mechatronics Engineering Sep 2020 – Jun 2024
National Polytechnic University of Armenia
GPA: 3.97/4.00 — Highest Honours (Red Diploma). Bachelor's thesis received maximum grade and Excellence distinction.
Robotic Hand Project • *Thesis DOI*